

Vector Analysis in GIS using QGIS

Exercise 1: Create Grid for Nepal

- Upload Administrative Boundary shapefile (e.g., districts or provinces).
- Create 0.01° x 0.01° Grid: Vector → Research Tools → Create Grid.
- Remove Non-Overlapping Grids: (Select by Locations - select feature and intersect → go to the attribute table and delete selected)

Exercise 2: Census Data Feature Extraction

- Upload Census Excel file: Layer → Add Layer → Add Layer from Delimited Text.
- Join with Admin Layer: Layer Properties → Joins.
- Create Population Ratio Field in Attribute Table using Field Calculator. Output Field Name: female_male_ratio
 *$\text{round} ("census_education_female" * 100.0 / "census_education_male", 2)$*
- Visualize as Pie Chart: Layer Properties → Diagrams → Pie Chart.

Exercise 3: Line-Polygon Intersection

- Upload Roads and Admin Layers.
- Perform Intersection: Vector → Geoprocessing Tools → Intersection.
- Calculate Length: Attribute Table → Field Calculator
 $\text{round}(\text{length}(\$geometry)/1000, 2)$
- Summarize by District: Vector → Analysis Tools → Statistics by Categories. Choose **length_km** as the field to summarize

Exercise 4: Rainfall Data Voronoi Analysis

- Import Rainfall shapefile.
- Create Voronoi Polygons: Vector → Geometry Tools → Voronoi Polygons
- Clip Voronoi polygons to Nepal using Clip tool.

Exercise 5: Nearest Neighbor Analysis & Visualization

- Upload Settlement shapefiles.
- Convert Multipart to Single points if necessary.
- Compute Nearest Distance: Vector → Analysis Tools → Distance Matrix with rainfall stations.
- Visualize: Graduated symbology for distances, label nearest airport names.

Exercise 6: Geoprocessing Tools – Buffer, Intersection & Union

Create Buffer:

- Uploads Roads layer and create buffer with 300m.
- Vector → Geoprocessing Tools → Buffer.
- Dissolve buffers if overlapping is not needed
- Save as a new layer: Roads_Buffer

Clip or Intersect:

- Uploads Land Use layer.
- Cut one layer by another (e.g., land cover inside buffer)
- Go to: Vector → Geoprocessing Tools → Clip
- Save the output as **LandUse_in_Buffer**

Use Clip/Intersection to extract spatial relationships.

- Union: Vector → Geoprocessing Tools → Union.
- Calculate Area of Intersection: Field Calculator → $\text{area}(\$geometry)/10000$.
- Visualize: Use symbology and labels to enhance interpretation.
- Export Map Layout: Project → New Print Layout → Add map elements → Export.

Union Multiple Layers

- Go to: Vector → Geoprocessing Tools → Union
- Combine two polygon layers (e.g., water bodies and protected areas)